

Additional Resources and Going Further

Project ideas

- [Practice using all of the basic peripherals](#)
- Recreate the [examples from the Arduino IDE](#)
- Projects don't need to be serious or even useful! Try building a
 - simple game,
 - a “fidget device” (something that produces interesting outputs based on fidgety inputs from a user), or
 - a “screensaver” (something that produces gently changing and semi-random outputs continuously).
- Learn how to use the [STM32CubeMonitor](#) program to create computer-based dashboards that interact with your USB-connected STM32.

Debugging

- Books
 - [Debugging](#) (David Agans)
 - [If I Only Changed the Software, Why is the Phone on Fire?](#) (Lisa Simone)
 - [Debugging by Thinking: A Multidisciplinary Approach](#) (Robert Metzger)
 - [Why Programs Fail: A Guide to Systematic Debugging](#) (Andreas Zeller)
 - [Debug It!: Find, Repair, and Prevent Bugs in Your Code](#) (Paul Butcher)
- Articles
 - [How to Debug](#) (John Regehr)
 - [Blogs tagged "debugging" from Embedded Artistry](#) (Phillip Johnston)
- Appnotes/Manuals
 - [UM2609: STM32CubeIDE user guide](#) (particularly sections 3.5 and 3.6)
 - [AN4989: STM32 microcontroller debug toolbox](#)

STM32

- Books
 - [Embedded Systems Fundamentals with Arm Cortex-M based Microcontrollers: A Practical Approach Nucleo-F091RC Edition](#) (Alexander Dean)
 - [Hands-On RTOS with Microcontrollers: Building real-time embedded systems using FreeRTOS, STM32 MCUs, and SEGGER debug tools](#) (Brian Amos)
- Videos/Blogs
 - [Getting Started with STM32 and Nucleo](#) (Shawn Hymel)
 - [Phil's Lab](#)
 - STM32 Videos (from the STM32CubeIDE “Information Center”)
 - [Embedded Wednesdays](#) (Andrei Chichak)

Embedded Systems

- Books
 - [Learning a New Microcontroller](#) (Steve Branam)
 - [Making Embedded Systems](#) (Elecia White)
 - [Embedded Systems Fundamentals with Arm Cortex-M based Microcontrollers: A Practical Approach Nucleo-F091RC Edition](#) (Alexander Dean)
 - [Test-Driven Development for Embedded C](#) (James Grenning)
 - [Design Patterns for Embedded Systems in C](#) (Bruce Douglass)
 - [Better Embedded System Software](#) (Philip Koopman)
 - [Practical UML Statecharts in C/C++](#) (Miro Samek)
 - [Hands-On RTOS with Microcontrollers: Building real-time embedded systems using FreeRTOS, STM32 MCUs, and SEGGER debug tools](#) (Brian Amos)
- Websites/Podcasts
 - [Embedded.fm](#)¹
 - [EmbeddedArtistry.com](#)²
 - [Interrupt Blog](#) from Memfault
 - [Embedded Online](#) and the [Embedded Online Conference](#)
- Articles
 - [Embedded for Everyone](#) (Nathan Jones)

¹ By supporting the Embedded.fm podcast on Patreon, you are invited to join a members-only Slack channel. Doing that was one of the best professional decisions I made in the last few years! The community has been extremely knowledgeable, helpful, and welcoming. I've learned so much from them!

² Embedded Artistry also has a members-only forum which has been tremendous! Phillip Johnston is a remarkably wise and intelligent individual.